

DeSales University Code Jam Manual

Code Jam Application Manual

CS- 453 | DeSales University, 2755 Station Avenue, Center Valley, PA

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Overview

Introduction

Welcome to the DeSales University Code Jam User Manual. This document is intended to assist admins and judges in using the system, as well as informing teams participating in the Code Jam about usage. New users will be provided with log-in credentials which can be later changed. The link to the Code Jam website can be found here: [DeSales University Code Jam Website](#).

Purpose

This system was developed to assist DeSales University in hosting their annual Code Jam contest. Our investor, Luke Cossman, has presented concerns about scoreboard visibility, submission automation, and other quality-of-life improvements that an application could resolve. This system provides an easy and intuitive method for teams to develop, test, and submit code and receive judge feedback. It also allows admins and judges a means of running submissions, viewing associated test cases, providing feedback, and ultimately scoring each submission. Lastly, a controls page is provided to assist in setup and resetting of database values for each Code Jam.

Brief Description

On the team-facing page, competitors can write and run code for provided problems. They can then submit their code, work on other problems, and await judge feedback for their submissions. On the judge-facing page, judges can select, view, and run submitted code, provide feedback, and score submissions. Judges also have access to the scoreboard for the event, which they can open in a separate instance for display at the event. A controls page is provided for judges and admins to generate passwords for teams, change their passwords, hide and show the scoreboard, and change database values.

Code Jam

Code Jam is an annual coding competition hosted by DeSales University. The competition consists of teams and judges. Each team must complete 10 problems and submit each for scoring. Feedback is provided at the judges' discretion, and scoring reflects the number of previously incorrect submissions and teams that submitted correct solutions beforehand. Judges run and score code based on stored test cases set before the event. Admins have the same responsibilities as judges but are also responsible for setting test cases before the event and clearing database values afterwards.

System Management

Docker

Code Jam is fully containerized using Docker. The entire application, runtime environment, dependencies, and database, is packaged and executed within Docker containers. This ensures the system runs consistently across different machines without requiring manual configuration of libraries or services. Docker manages all required components of the application. This eliminates the need to install or configure these components individually on the host machine.

Persistent data, such as generated team credentials and coding submissions, is stored using Docker volumes. These volumes allow data to remain intact even if containers are stopped, restarted, or rebuilt. The containers remain active while the application is in use and can be started or stopped as needed through Docker commands.

Docker is the only required installation. All other dependencies are handled automatically within the containerized environment.

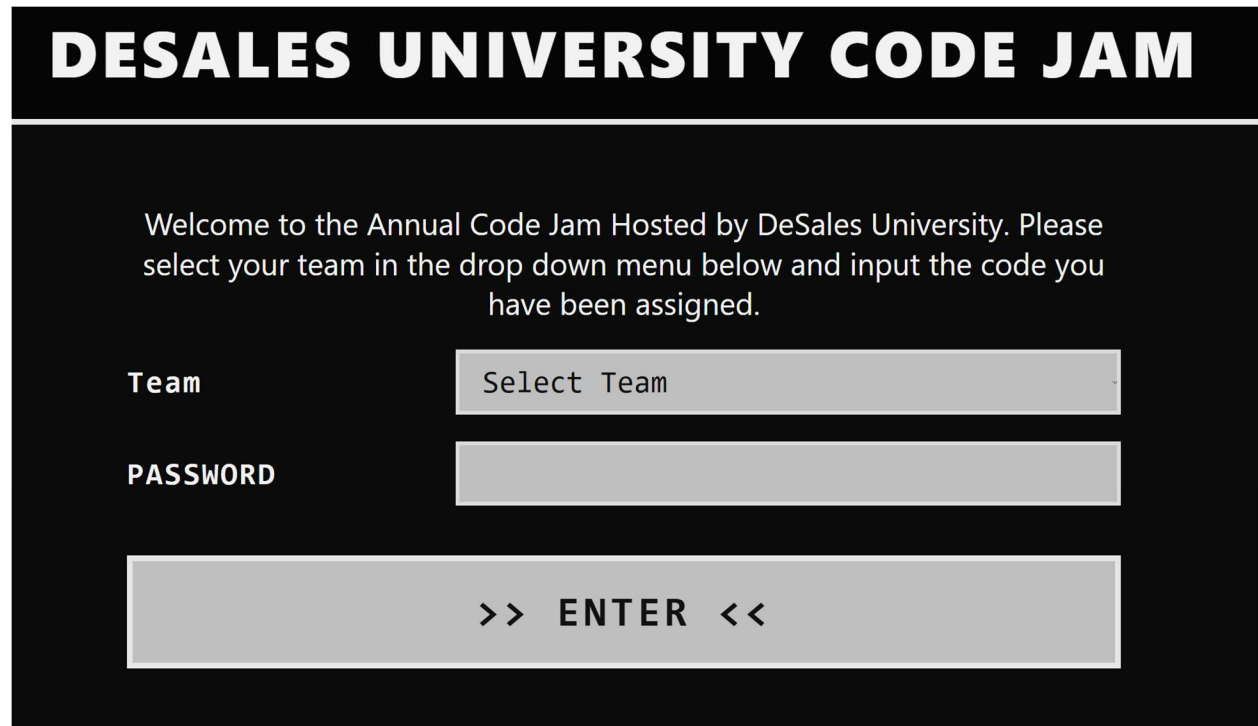
Program Modifications

For all program modifications and changes, docker must be composed down and reloaded to accept new changes.

1 Getting Started

1.1 Login Page & Account Selection

Judges and teams access their designated pages through the main login page. On the page is a drop-down menu presenting all available teams and judge accounts. The passwords for team accounts will be sent directly to the teams from DeSales University judges.



DESALES UNIVERSITY CODE JAM

Welcome to the Annual Code Jam Hosted by DeSales University. Please select your team in the drop down menu below and input the code you have been assigned.

Team

PASSWORD

>> ENTER <<

Figure 1: Login Page

1.2 Setting up Code Jam

To set up a Code Jam event, an admin must log into the system. The first step is to navigate to the Controls page and clear the database using the Clear Contest Data to ensure any remaining values from the last event are removed. After deciding what problems will be 1-10 and deciding the test cases for each, utilize the Update Test Cases button to set the test case values for the current event. Once it's decided how many teams will be participating (max of 20), use the team generator to create team name and password pairings. It is recommended that 20 be used for each event, as if more teams are needed, generating more teams removes all old data from teams table. Lastly, team names and passwords must be individually distributed to users at the time of the event, so teams can access their assigned pages.

2 Judge Facing Interface

The role of judges is to review and score each of the team's submissions. For each submission, the judges will provide feedback to the team along with a correct/incorrect verdict. Judges maintain the scoreboard and team passwords from the controls interface.

2.2 Submission

The primary judge interface includes an IDE, terminal, test cases, and a feedback window. The Submission Code window displays the team's code. Below it, an interactive terminal shows the program's output. For each problem, judges see the test case inputs and expected outputs. Problems can be viewed individually or as a list of all submissions. Selecting a problem shows only that team's submission.

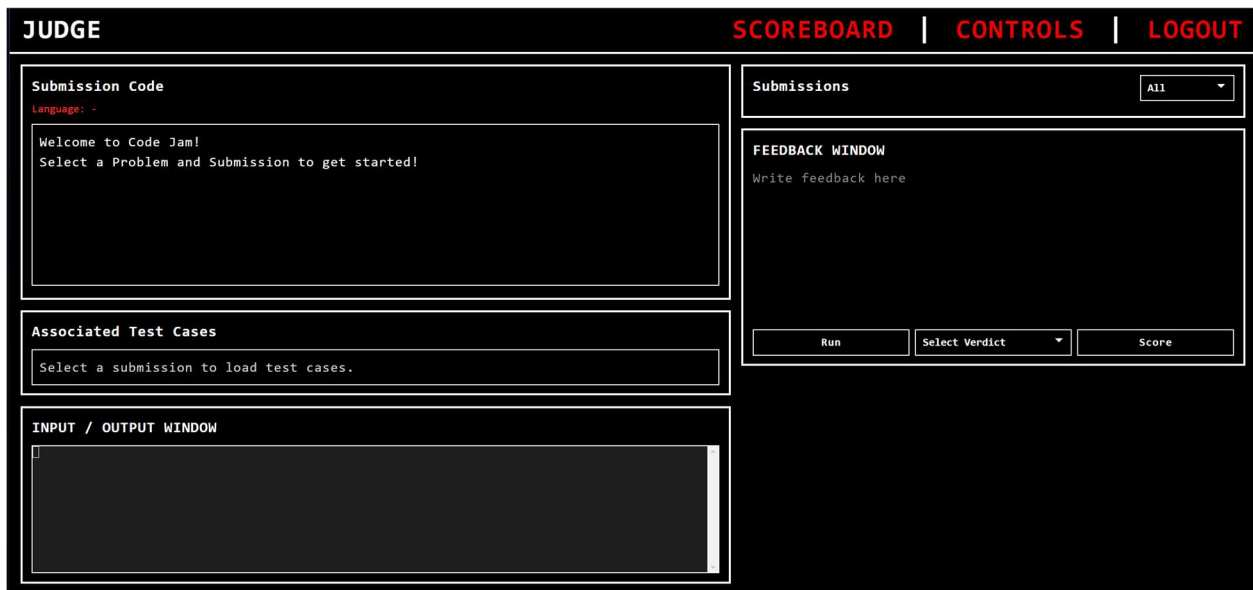


Figure 2: Judge Facing Submission Page

After selecting a problem, the judge can mark it as correct or incorrect based on the results. A feedback window allows judges to send comments to the team.

2.2.1 Score Submission

Scoring is completed automatically. Each team starts with 20 points per problem. The score decreases by 1 for every other team who submits a correct solution and for each incorrect submission.

$$\text{Total Score} = 20 - \{\# \text{ of correct submissions by other teams}\} - \{\# \text{ of incorrect submissions}\}$$

2.3 Controls

The screenshot shows the 'ADMIN OPTIONS/DEBUG' section of a web interface. At the top right, there are links for '<< BACK TO JUDGE' and 'SCOREBOARD'. The main area is divided into two columns. The left column contains three sections: 1. 'NUMBER OF TEAMS' with a text input set to '0' and a 'Generate Teams' button. Below this, a green message says 'Generated 5 team account(s)'. 2. 'GENERATED TEAM ACCOUNTS' with a table listing five teams and their hashed passwords. 3. 'CHANGE ACCOUNT PASSWORD' with a dropdown for account selection, a new password input, and a 'Change Password' button. The right column contains three large buttons: 'HIDE SCOREBOARD', 'UPDATE TEST CASES', and 'CLEAR CONTEST DATA'.

USERNAME	PASSWORD
Team1	Uaa&Hb%YhsoG
Team2	dkZGwP39VQkh
Team3	9JcW&jq#EyR4
Team4	Wu%&IdVu3MWP
Team5	FzsJfemIy#j

Figure 3: Judge Control Page

2.3.1 Generate Teams

Judges can generate up to 20 teams with randomized passwords which are hashed into the database. Generating new teams overwrites previous passwords to maintain account security and prevent access by past participants.

To generate team passwords, enter the number of teams desired and then click “Generate Teams”. The generated teams and passwords will be displayed below the control. Passwords are stored as hashed values but presented as plain text values to send to teams.

2.3.2 Change Judge Passwords

Judge passwords may also be changed via the “Change Account Password” window. Only the head admin may be permitted to change judge passwords.

2.3.3 Update Test Cases

Test cases may be updated or changed at any time. To do so, a judge may select the “Update Test Cases” button. The Code Jam program will then ask the judge for which problem they would like to update as well as the input and expected output of test case. Both input text and expected output will be updated and will be visible as the submissions arrive in the submission page of the judge facing.

The screenshot shows a modal dialog titled 'Update Test Cases'. It contains four input fields: a dropdown for 'Problem' with the placeholder '-- Select problem --', a text input for 'Test Case Number' with the value '0', a text area for 'Input Text', and another text area for 'Expected Output'. At the bottom right, there are two buttons: 'Cancel' and 'Save Test Case'.

Figure 4: Judge Update Test Cases

2.3.4 Clear Contest Data

At the end of Code Jam, judges may clear all student submissions. The clearing of submissions is to prevent future contestants from utilizing code from previous years.

2.4 Scoreboard

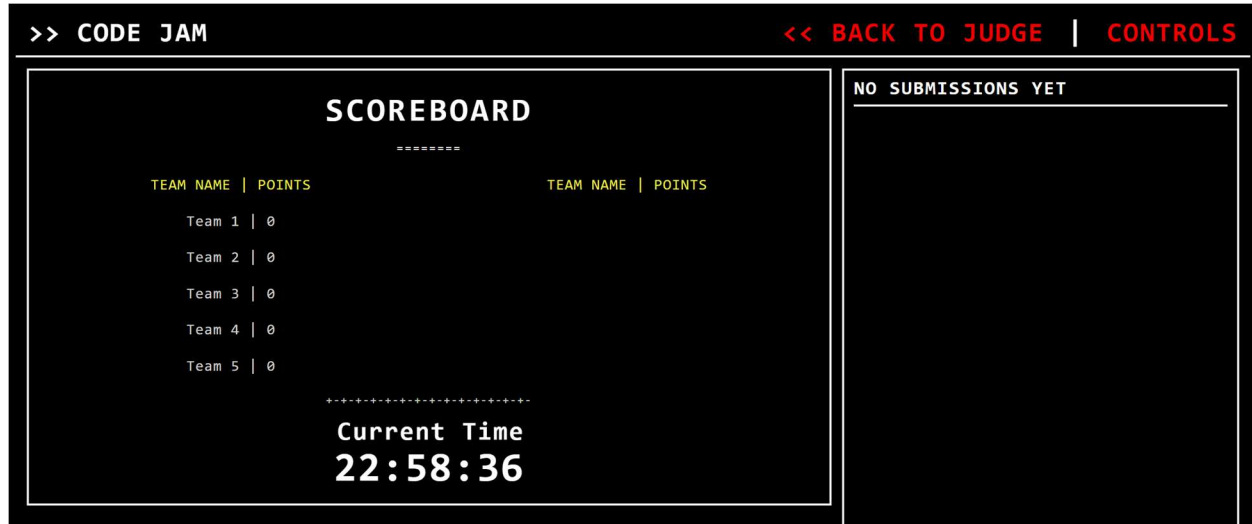


Figure 5: Scoreboard

The scoreboard presents a live update of all the team's scores throughout the Code Jam. As teams receive feedback that their solution code is correct, the scoreboard will update with the number of points they earned.

The scoreboard also displays submissions as they arrive. Contestants may look at the scoreboard at any point in time during the Code Jam; especially after submitting code for feedback. This display will serve as confirmation that their code has been received by judges.

2.4.1 Scoreboard Visibility

Towards the final 30 minutes of Code Jam, judges may hide the scoreboard, via judge controls, to prevent teams from guessing which team will place in the finals. To hide the scoreboard, a judge will navigate to controls and then select "Hide Scoreboard." In the event of an accident hiding, a judge may unhide the scoreboard utilizing the same button.

3 Team Facing Interface

For the Code Jam event, teams can access a built-in IDE to run and test their cases for the problems assigned. The page consists of arrows to switch between problems, a feedback section to view the current problem selected, whether a judge has submitted feedback, submission status, and points awarded. Below the IDE are the Run and Submit buttons, which saves and submits the code written in the IDE.

3.1 Team IDE

Inside the IDE, users can switch between languages, and if saved, each problem can display their respective code written. The IDE supports 4 languages currently, each with syntax highlighting, function name sensing, indentation, etc.

The terminal displays code errors when code is run, highlighting the line and specific error for users to change

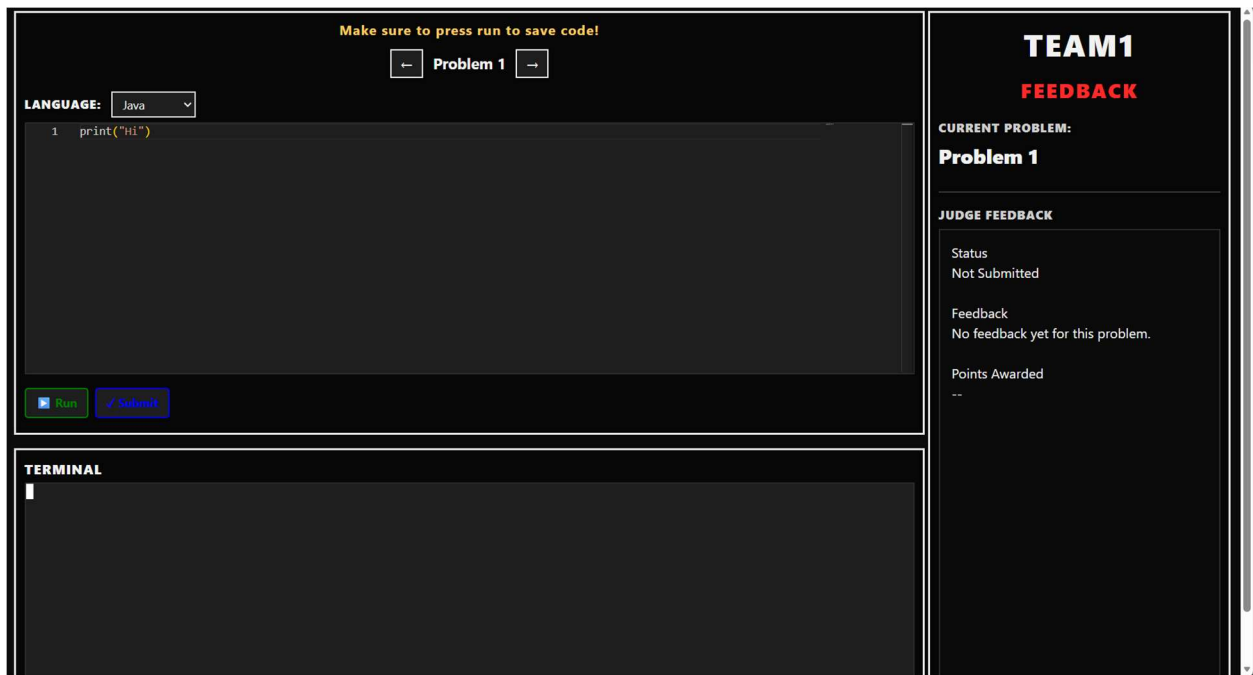


Figure 6: Team IDE

3.1.1 Supported Languages

The languages supported are Java, JavaScript, Python and C++

3.2 Judge Feedback & Scoring

Judges can submit feedback in real time to submitted answers by teams. Team contestants can view whether an answer submitted is labeled correctly or incorrect, any feedback written by judges, and their score.

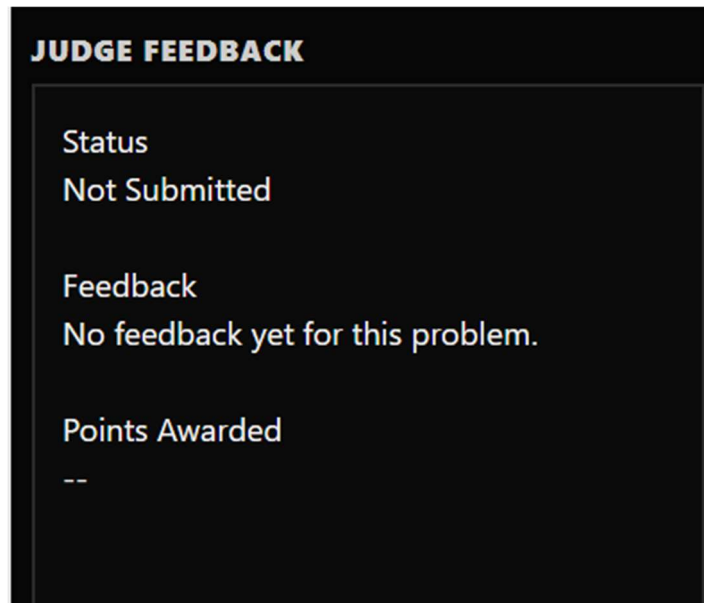


Figure 7: Team Feedback from Judge